GenMatter: Perceiving Physical Objects with Generative Matter Models

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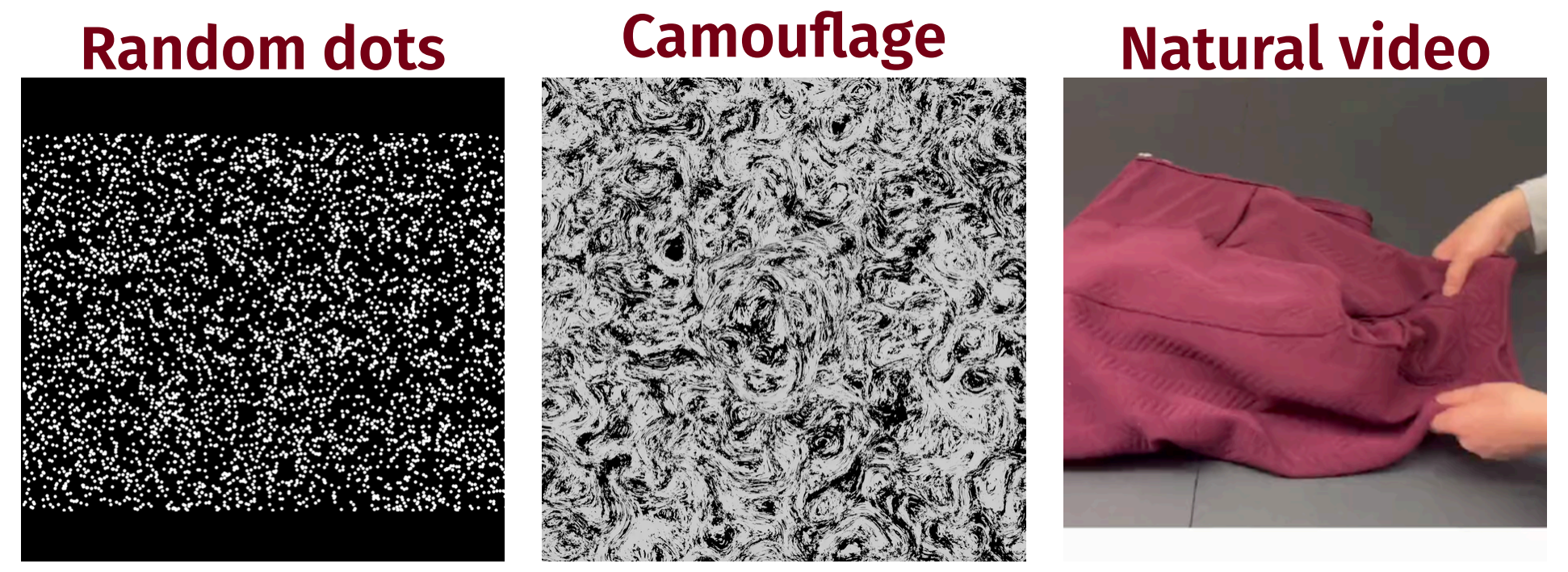
Project page & videos

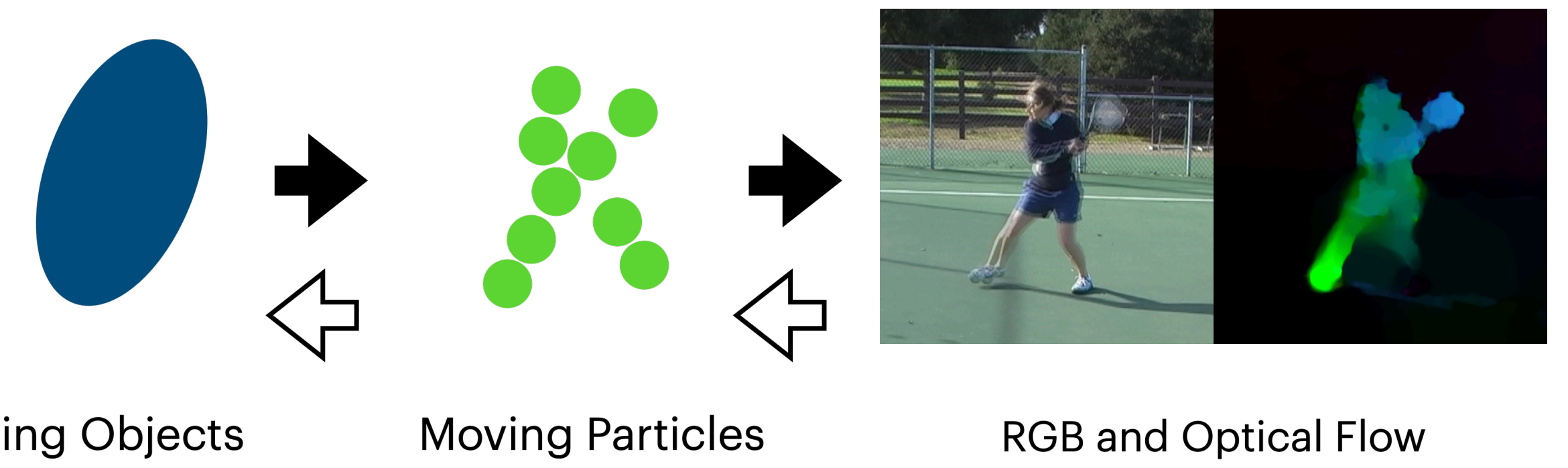
Abstract

GenMatter frames perception as **Bayesian inference** in a generative model of matter: a scene is particles (small Gaussians of local matter) grouped into clusters (independently moveable objects), recovered **without learning**, through probabilistic inference. **One model** spans random dots, camouflaged textures, and natural video, where biological vision succeeds but computer vision does not.

One visual system, three worlds



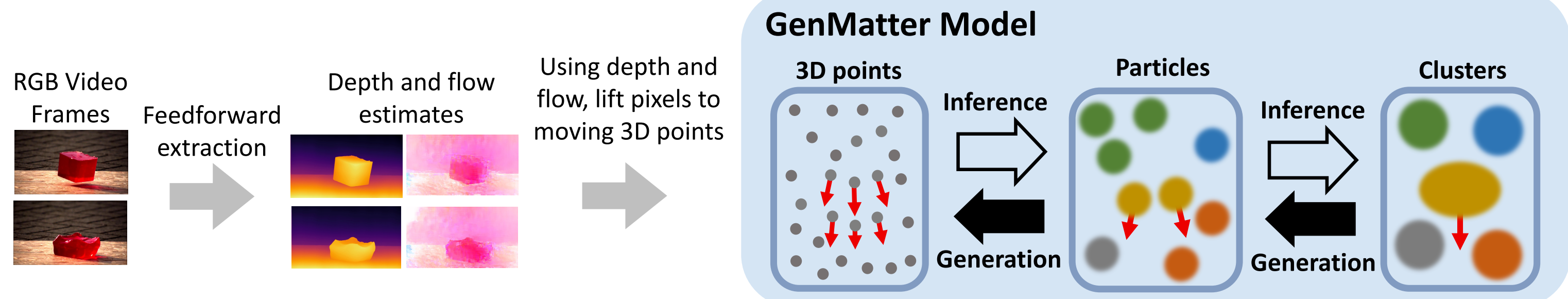
Three worlds, one model: random dots, camouflage, and natural video. Humans solve all three, but machines are not as general.



One model, run both ways: forward, it renders a scene (solid arrows), and inverted, it perceives the objects (hollow arrows).

- **Motion comes in rigid pieces:** each object paints its own patch of optical flow, which grouping segments into objects.
- **Domain-general by construction:** the same model perceives all three worlds, with no per-task training or hand-crafted regularizers.

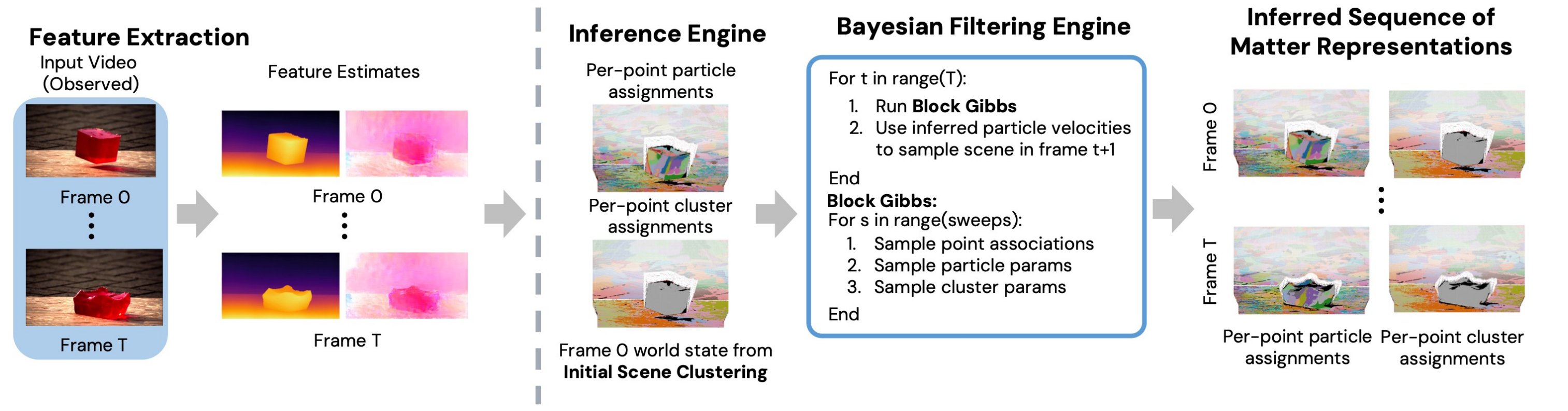
The generative matter model



Depth and flow lift every pixel to a moving 3D point, and perception inverts that generative process.

- **Two-level hierarchy with slack:** an object is a cluster of particles, each free to deviate from the cluster's rigid motion, so one model fits both rigid and deforming objects.
- **Perception is inference:** GenMatter infers the hidden objects and their motions, grouping the particles that move as one.

Inference

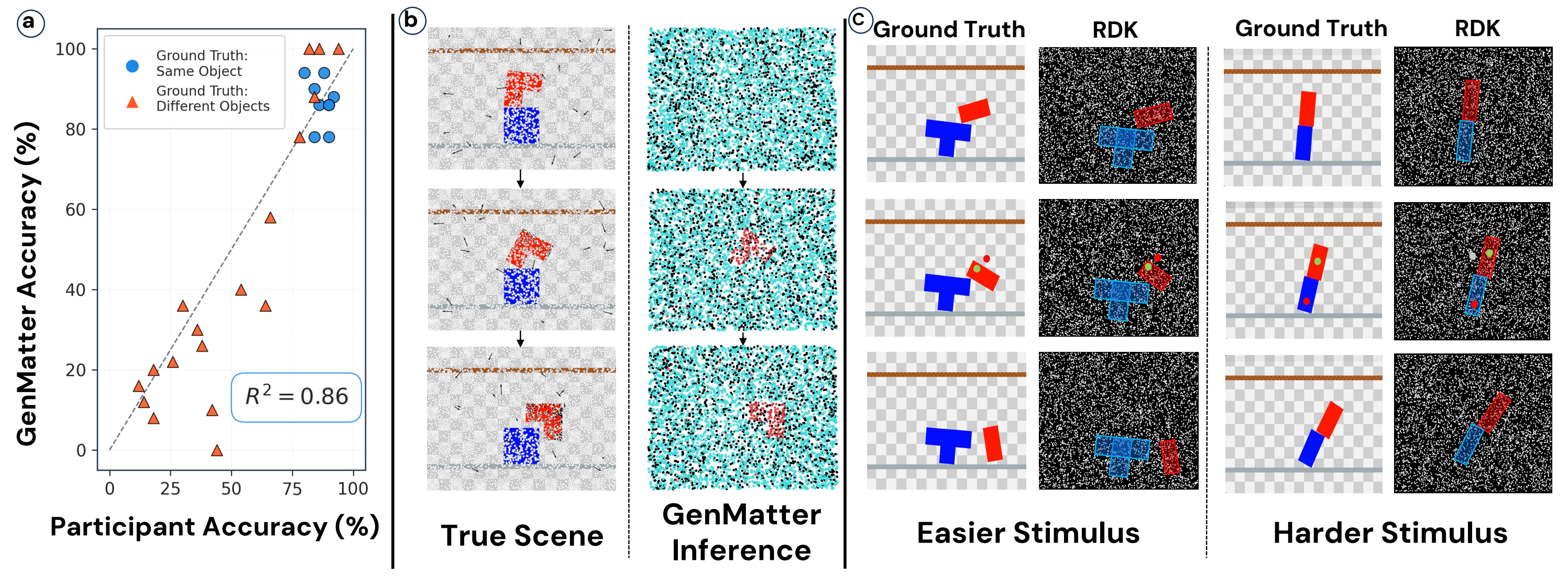

Feature Extraction
Input Video (Observed)
Frame 0
Frame T
Feature Estimates
Frame 0
Frame T
Inference Engine
Per-point particle assignments
Per-point cluster assignments
Frame 0 world state from Initial Scene Clustering
Bayesian Filtering Engine
Inferred Sequence of Matter Representations
Frame 0
Frame T
Per-point particle assignments
Per-point cluster assignments

```
For t in range(T):
    1. Run Block Gibbs
    2. Use inferred particle velocities
       to sample scene in frame t+1
End
Block Gibbs:
For s in range(sweeps):
    1. Sample point associations
    2. Sample particle params
    3. Sample cluster params
End
```

A training-free inference engine: feature extraction, then block-Gibbs sampling, then Bayesian filtering.

- **Structure makes inference tractable:** conditional independence across the hierarchy lets GenMatter sample every level in parallel, so posterior inference runs online, a sweep or two per frame.
- **Deep features for structured inference:** pretrained depth, flow, and DINO networks extract the features, and GenMatter's inference groups the points into objects.
- **Training-free core:** with no task-specific finetuning, a vectorized GenJAX engine runs on one L4 GPU in a few kilobytes of code.

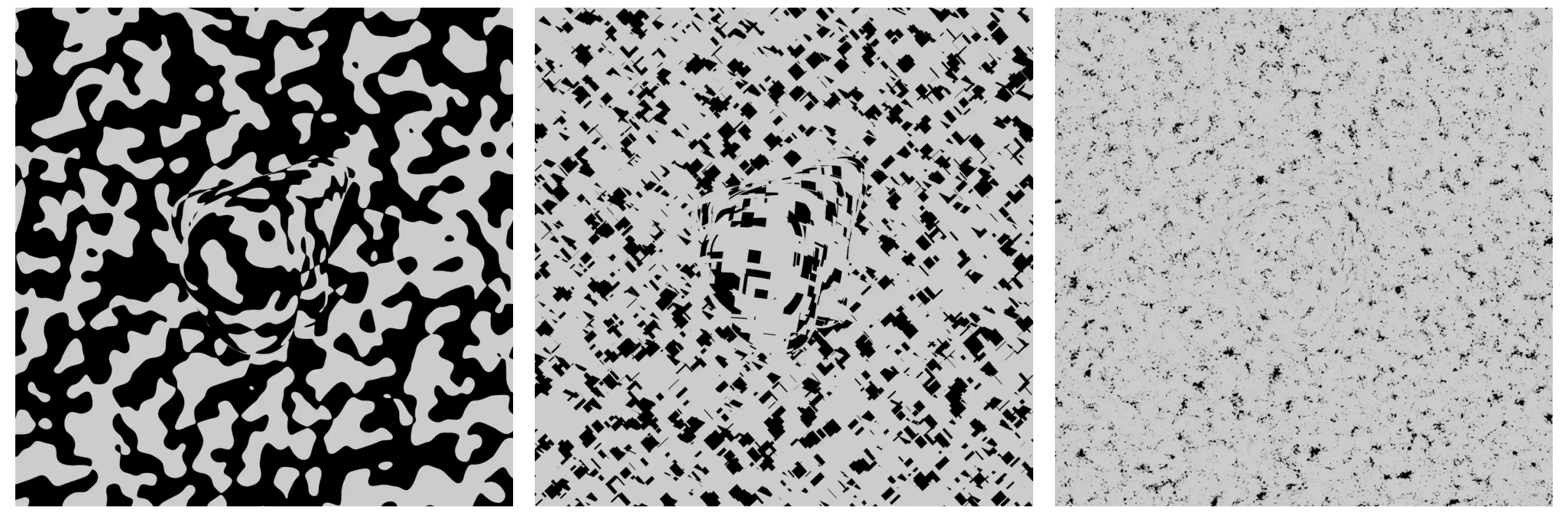
① Human object perception



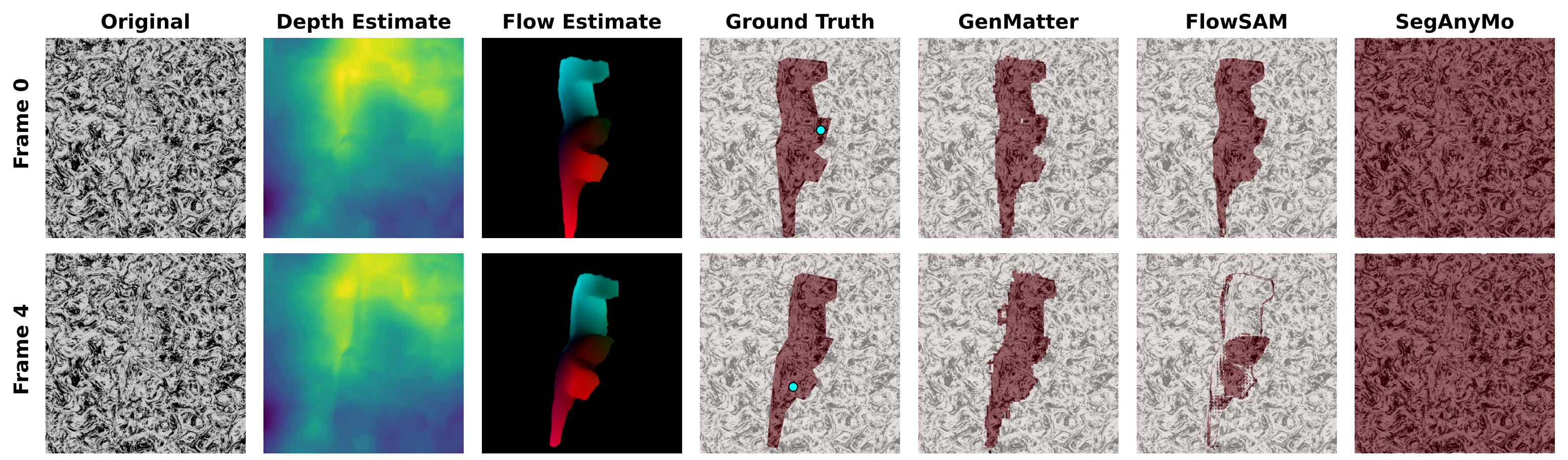
150 participants judged whether two probe dots share an object across 27 kinematograms. GenMatter matches them at $r^2 = 0.86$.

- **Cluster grouping is essential:** without the cluster layer, 5- and 500-particle variants reach only $r^2 = 0.35$ / 0.41, with learned FlowSAM at $r^2 = 0.04$, far below the full model.
- **Captures graded uncertainty:** the model's confidence tracks how often people group the dots, staying calibrated even on the hardest, most ambiguous trials.

② Structure from motion in camouflage



Find the moving object: each panel is a single texture frame, where only motion reveals the hidden 3D object.

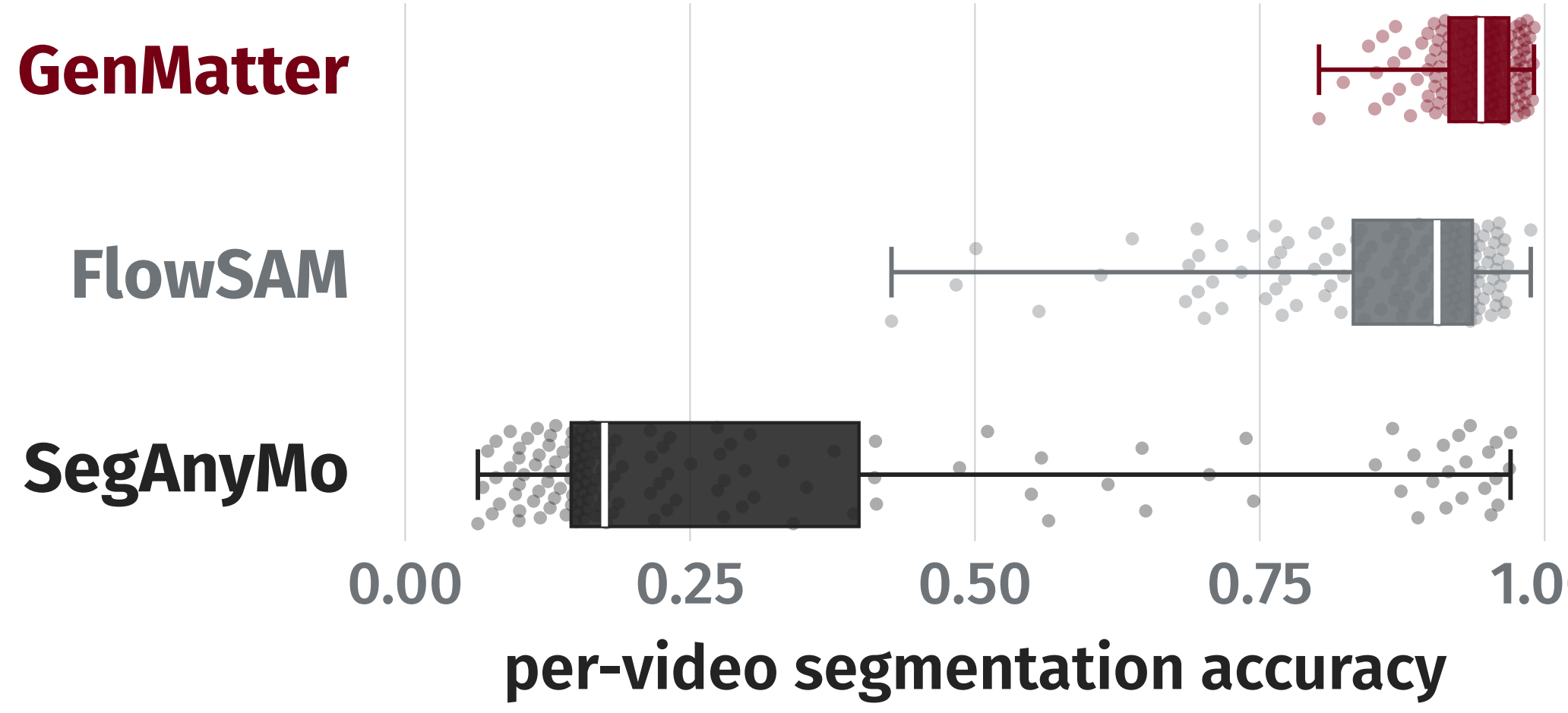


Motion reveals 3D shape. GenMatter recovers the hidden shape and mask where learned baselines drift or fail.

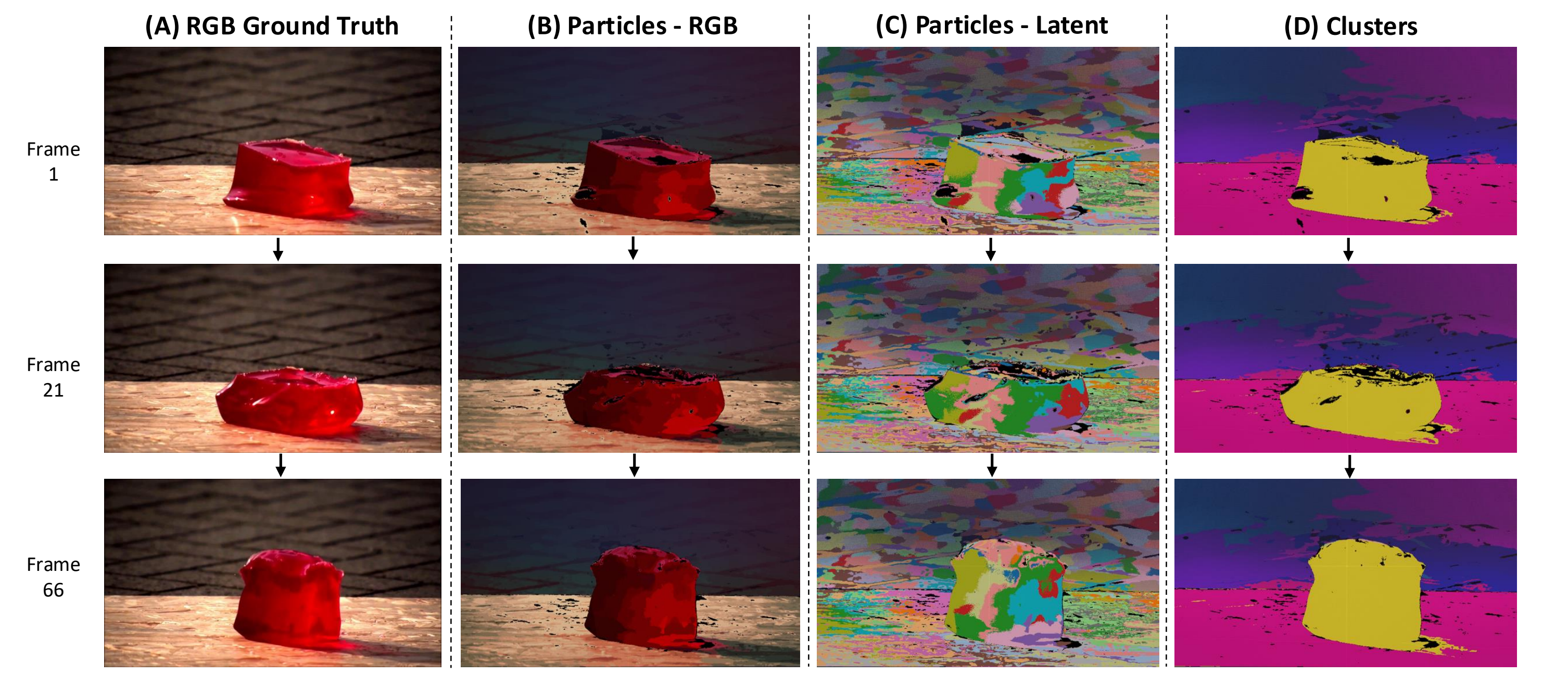
- **Structure from motion, not appearance:** GenMatter recasts the Gestalt principle of common fate as Bayesian inference, finding objects that no appearance-based feedforward model can see.
- **Outperforms the learned baselines:** higher Jaccard than FlowSAM on 111 of 140 videos and SegAnyMo on 133 of 140, even with depth ablated and motion as the only cue.

Metric	SegAnyMo	FlowSAM	GenMatter
Accuracy	0.33 [.28,.37]	0.87 [.85,.88]	**0.94** [.93,.94]
Jaccard	0.26 [.22,.31]	0.67 [.63,.70]	**0.72** [.70,.74]

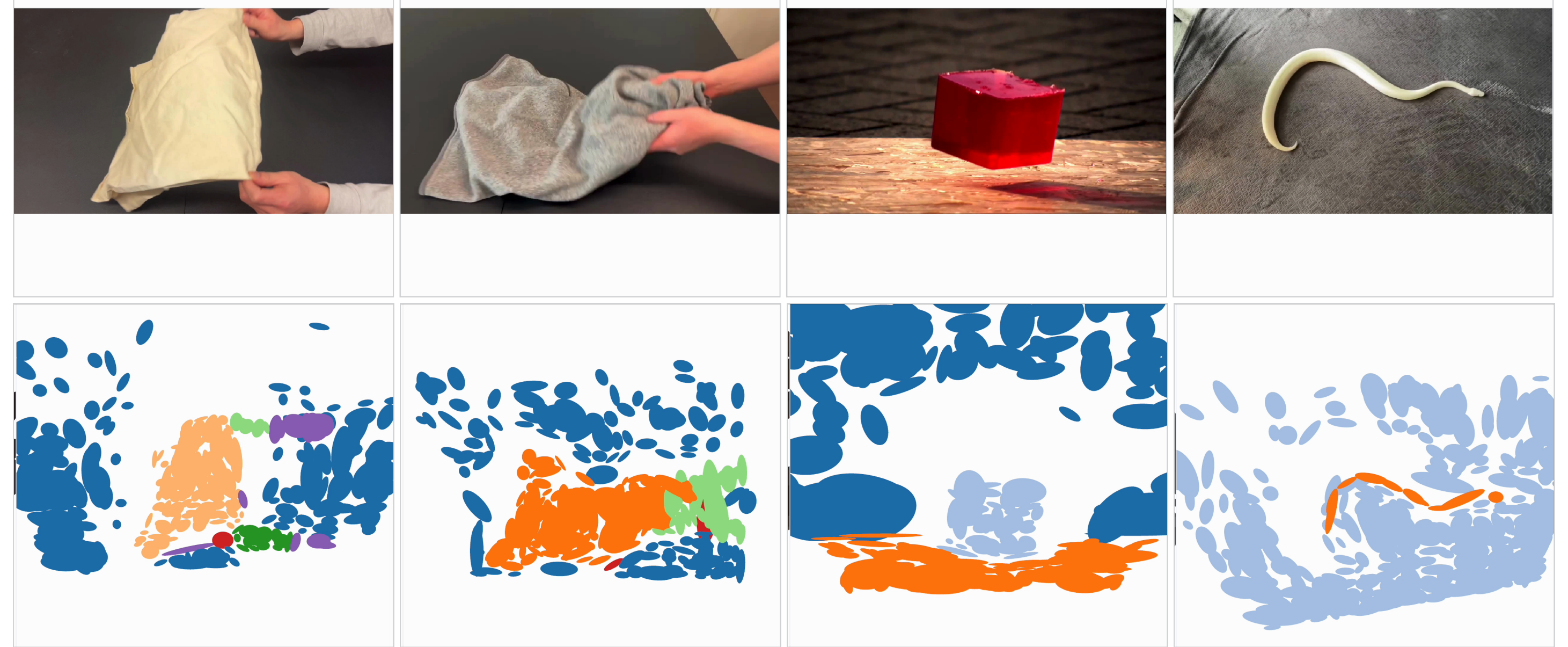
Mean accuracy and Jaccard [95% bootstrap CI]: GenMatter is highest and tightest on both.



③ 3D particles from natural video



A deforming jello cube: velocity slack lets the particles follow the bits of jello deformed by the collision, while the cluster stays one object.



Deformable, weakly textured objects: input frames (top) and the inferred object clusters (bottom).

Metric	CoTracker3	GenMatter	GenMatter (abl.)
J_m (SAM)	0.78 [.69,.87]	**0.79** [.73,.84]	0.69 [.61,.77]
J_m (GT)	0.78 [.69,.87]	**0.77** [.73,.84]	0.68 [.58,.73]

Jaccard J_m on TAP-Vid-DAVIS [95% bootstrap CI]. The ablation (abl.) drops the cluster hierarchy and falls well short.

- **Matches a supervised tracker, training-free:** projecting inferred 3D particles to 2D reaches CoTracker3's DAVIS accuracy, and its probabilistic model exposes a speed-accuracy design space, up to 12× faster by subsampling.

Conclusion

- **One model, many viewing conditions:** a single probabilistic engine spans random dots, camouflage, and natural video, approaching the generality of biological vision that no single learning system has matched.
- **Bridges human vision and computer vision:** object-level scene understanding emerges from structured inference.